

Assignment

August 10, 2026

1 Examples of Sets in Set-Builder Form

Even numbers

$$A = \{x \in \mathbb{Z} \mid x \text{ is even}\}$$

Odd numbers

$$B = \{x \in \mathbb{Z} \mid x \text{ is odd}\}$$

Natural numbers less than 10

$$C = \{x \in \mathbb{N} \mid x < 10\}$$

Multiples of 5

$$D = \{x \in \mathbb{Z} \mid x = 5n, n \in \mathbb{Z}\}$$

Prime numbers

$$E = \{x \in \mathbb{N} \mid x \text{ is prime}\}$$

Integers between -5 and 5

$$F = \{x \in \mathbb{Z} \mid -5 \leq x \leq 5\}$$

Perfect squares

$$G = \{x \in \mathbb{N} \mid x = n^2, n \in \mathbb{N}\}$$

Real numbers greater than 0

$$H = \{x \in \mathbb{R} \mid x > 0\}$$

Students scoring above 80

$$I = \{x \in S \mid \text{score}(x) > 80\}$$

(where S is the set of all students)

Letters in the word APPLE

$$J = \{x \in \{A, B, C, \dots, Z\} \mid x \text{ appears in "APPLE"}\}$$

2 Inclusion Exclusion Principle

General Formula

Let $\{A_i\}_{i=1}^{\infty}$ be a countable collection of sets. Then:

$$\left| \bigcup_{i=1}^{\infty} A_i \right| = \sum_{k=1}^{\infty} (-1)^{k+1} \left(\sum_{1 \leq i_1 < i_2 < \dots < i_k} |A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}| \right)$$

This is the infinite extension of the inclusion–exclusion principle.

2.1 Proof (Sketch)

1. Finite Truncation

For any finite n , the inclusion–exclusion formula holds:

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} |A_{i_1} \cap \dots \cap A_{i_k}|$$

2. Limit Process

As $n \rightarrow \infty$, the union $\bigcup_{i=1}^n A_i$ approaches $\bigcup_{i=1}^{\infty} A_i$. If the series of intersection terms converges absolutely, the infinite alternating sum is valid.

3. Element-wise Counting

For any element x belonging to exactly m sets, its contribution is:

$$\binom{m}{1} - \binom{m}{2} + \binom{m}{3} - \dots + (-1)^{m+1} \binom{m}{m} = 1$$

This identity holds even when m is infinite, provided the alternating series converges.

4. Conclusion

Each element is counted exactly once, so the formula holds for infinite unions under convergence conditions.

3 Types of Sets with Examples

Type of Set	Definition	Example in AI/DS
Numerical Set	Contains quantitative values (discrete or continuous).	Sensor readings for IoT devices: {23.1, 24.5, 25.0, 26.2}
Categorical Set	Represents discrete labels or groups.	Sentiment labels: {Positive, Negative, Neutral}
Time Series Set	Ordered data indexed by time.	Stock prices: $\{(t_1, 100), (t_2, 102), (t_3, 98)\}$
Ordinal Set	Ranked values where order matters but not magnitude.	Customer satisfaction: {Poor, Fair, Good, Excellent}
Image Set	Collection of images for vision tasks.	MNIST dataset: handwritten digits {0, 1, 2, ..., 9}
Text Set	Corpus of documents or sentences.	NLP training data: {"AI is powerful", "Data drives decisions"}
Web/API Set	Data collected from online sources.	Tweets for sentiment analysis: {Tweet ₁ , Tweet ₂ , ...}
Multivariate Set	Involves multiple variables simultaneously.	Patient health records: {(Age, BP, Cholesterol)}
Correlation Set	Shows relationships between variables.	Study hours vs. exam scores: {(2, 60), (4, 75), (6, 90)}
File-based Set	Stored in structured formats like CSV/Excel.	Training dataset in CSV: {(Name, Age, Score)}

4 Partially Ordered Sets (Posets)

A **partially ordered set (poset)** is a set P together with a relation $\leq$ that satisfies:

- Reflexive: $a \leq a$ for all $a \in P$.
- Antisymmetric: If $a \leq b$ and $b \leq a$, then $a = b$.
- Transitive: If $a \leq b$ and $b \leq c$, then $a \leq c$.

Not all elements need to be comparable. For example, in the subset relation, $\{a\}$ and $\{b\}$ are not comparable.

4.1 Examples with Hasse Diagrams

1. Divisibility on $\{1, 2, 4\}$

Relation: $a \leq b$ if a divides b .

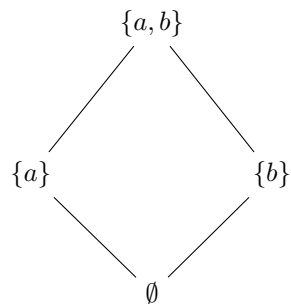
$$1 \leq 2 \leq 4$$



2. Power set of $\{a, b\}$

Relation: $\subseteq$ (subset relation).

$$\emptyset \subseteq \{a\}, \{b\} \subseteq \{a, b\}$$



3. Divisibility on $\{1, 2, 3, 6\}$

Relation: $a \leq b$ if a divides b .

$$1 \leq 2 \leq 6, \quad 1 \leq 3 \leq 6$$

